

SMARTBUDGET

Test Cases and Traceability Update

*19 functional, non-functional and security test cases, with SRS traceability — accompanies the
Software Test Plan*

Version 1.1

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Revision History

Date	Version	Description	Author
17 Sep 2026	1.0	Initial formal document derived from implemented SmartBudget prototype and project source archive	Amrutha, Akanksha, Aniketh
27 Sep 2026	1.1	Corrected traceability gaps between requirements and test cases; aligned NFR/SR coverage with actual test case content; corrected test case wording	Amrutha, Akanksha, Aniketh

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1. Purpose

The faculty requirement specifies a minimum of 7–10 test cases covering both functional and non-functional requirements, with security validation included. This document provides 19 test cases (TC-01 to TC-19) covering functional, non-functional, validation and security scenarios, together with a traceability matrix to the Software Requirements Specification and an execution record to be completed during test execution. Traceability is stated at the level of individual requirement IDs rather than requirement ranges, so that coverage claims match exactly what each test case verifies; two requirements (NFR-07, NFR-10) are implementation-level or presentation-level properties not exercised by a dedicated test case, and are flagged as such rather than force-mapped to an unrelated case.

2. Detailed Test Cases

ID	Type	SRS Ref.	Test Objective / Input	Expected Result
TC-01	Functional	FR-01	Valid unique signup	201 Created; account created; response JSON does not contain password_hash.
TC-02	Validation	FR-02	Signup with 5-character password	400 Bad Request; password-policy error; no account created.
TC-03	Security	FR-03 / SR-01	Repeat signup with same email; inspect stored password	409 Conflict on duplicate email; stored value is a bcrypt hash, not plaintext.
TC-04	Security	FR-04 / SR-04	Login with correct password, then with wrong password	Correct password succeeds (200); wrong password returns 401.
TC-05	Functional	FR-05	Create a Food expense of ₹450	201 Created; transaction saved; frontend displays the expense as negative.
TC-06	Validation	FR-05 / SR-05	Create transaction with amount = “abc”	400 Bad Request; transaction not created.
TC-07	Functional	FR-06–FR-07	Create, list, then delete a transaction	Record appears in the user’s list; delete succeeds and removes it.
TC-08	Validation	FR-08 / SR-06	Set budget to 0, then a positive value, then a negative value	0 and positive values save; negative value returns 400.
TC-09	Functional	FR-09–FR-10	Import supplied sample_bank_statement.csv	Rows import successfully; recognised narrations map to expected categories; unrecognised narrations map to Other.
TC-10	Functional	FR-11	Create transactions across multiple categories and months	Analytics totals correctly reflect the underlying transaction data.
TC-11	Functional	FR-13	Export the Reports view as CSV	A CSV file is generated containing the expected report/transaction fields.
TC-12	Validation	FR-12 / SR-07	Create a goal with target 10000; attempt to create a second goal with target 0; then delete the valid goal	Positive target creates the goal; zero target returns 400 and is not created; deletion of the valid goal succeeds and removes it.

ID	Type	SRS Ref.	Test Objective / Input	Expected Result
TC-13	Functional	FR-14	Toggle theme, then refresh the page	Selected theme persists across page refresh (localStorage).
TC-14	Functional	FR-15	Click “Detect” on the auto-detection card	A fixed simulated transaction is inserted into the client-side transaction list.
TC-15	Non-functional	NFR-01	Open all primary navigation destinations	All six destinations are reachable as separate screens.
TC-16	Non-functional	NFR-05	Trigger expected API error conditions (invalid input, missing resource)	Appropriate HTTP status codes and JSON error messages are returned.
TC-17	Security	NFR-08	Create records for two different users; query each user’s data	Each user’s query returns only that user’s own records (no cross-user leakage).
TC-18	Non-functional	NFR-09	Run the frontend build and start the backend server	Frontend build completes without error; backend starts successfully with MongoDB connected.
TC-19	Non-functional (inspection)	NFR-06	Inspect the project source tree for the client, API-route and persistence code	Presentation code (React), routing/business logic (Express routes) and persistence code (Mongoose models) reside in distinct source modules with no direct cross-layer access.

Preconditions common to all cases: the MongoDB instance is running and reachable; the Express API is started on port 5001; the frontend is served through the Vite development server. Test data for TC-09 is the supplied sample_bank_statement.csv file.

3. Traceability Matrix

SRS Requirement	Test Case(s)	Coverage
FR-01	TC-01	Account creation
FR-02	TC-02	Signup validation
FR-03	TC-03	Duplicate email
FR-04	TC-04	Login
FR-05	TC-05, TC-06	Create + amount validation
FR-06–FR-07	TC-07	Read / delete
FR-08	TC-08	Budget settings
FR-09–FR-10	TC-09	CSV import + categorization
FR-11	TC-10	Analytics
FR-12	TC-12	Goals
FR-13	TC-11	Report export
FR-14	TC-13	Theme
FR-15	TC-14	Auto-detection

SRS Requirement	Test Case(s)	Coverage
NFR-01	TC-15	Navigation reachability
NFR-02	TC-06, TC-08, TC-12	Explicit validation errors on invalid numeric input
NFR-03	TC-03	Passwords stored as bcrypt hash
NFR-04	TC-01	password_hash excluded from serialized response
NFR-05	TC-16	API error status codes / JSON error messages
NFR-06	TC-19	Layer separation (verified by source inspection, not runtime test)
NFR-07	Not covered by a dedicated test case	Asynchronous backend operations — implementation-level property; recommend code review rather than a functional test
NFR-08	TC-17	Per-user data isolation
NFR-09	TC-18	Build / runtime via Node/Vite scripts
NFR-10	Not covered by a dedicated test case	Text-label conveyance of state/category — recommend a manual UI/accessibility review
SR-01	TC-03	bcrypt hashing before persistence
SR-02	TC-01	password_hash removed from serialization
SR-03	TC-02	Password length policy
SR-04	TC-04	bcrypt login verification
SR-05	TC-06	Reject non-numeric transaction amount
SR-06	TC-08	Budget numeric and ≥ 0
SR-07	TC-12	Goal target numeric and > 0
SR-08	Test Plan §5.1 (ID validation check)	ObjectId syntax validated before deletion — verified in the Test Plan security validation table; no dedicated TC ID

4. Execution Record

To be completed by the tester during execution. Status values: Pass, Fail, Blocked, Not Executed.

TC ID	Actual Result	Status	Tester	Date	Defect / Remarks
TC-01		Not Executed			
TC-02		Not Executed			
TC-03		Not Executed			
TC-04		Not Executed			
TC-05		Not Executed			
TC-06		Not Executed			
TC-07		Not Executed			

TC ID	Actual Result	Status	Tester	Date	Defect / Remarks
TC-08		Not Executed			
TC-09		Not Executed			
TC-10		Not Executed			
TC-11		Not Executed			
TC-12		Not Executed			
TC-13		Not Executed			
TC-14		Not Executed			
TC-15		Not Executed			
TC-16		Not Executed			
TC-17		Not Executed			
TC-18		Not Executed			
TC-19		Not Executed			